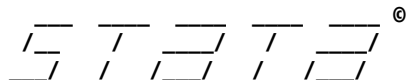


User: result to print



17.0
MP-Parallel Edition

Statistics and Data Science

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Notes:

1. Unicode is supported; see [help unicode advice](#).
2. More than 2 billion observations are allowed; see [help obs advice](#).
3. Maximum number of variables is set to 5,000; see [help set maxvar](#).

1 . import excel "C:\Users\hp\Downloads\Macro Project (2).xlsx", firstrow clear
(24 vars, 1,000 obs)

2 . drop M N O P Q R S T U V W X

3 . encode STATE, gen(state_id)

4 . xtset state_id YEAR

Panel variable: **state_id** (unbalanced)
Time variable: **YEAR, 2011 to 2024**
Delta: **1 unit**

5 . rename StatewiseCreditDepositRatio Bank

6 . rename STATEWISEAVAILABILITYOFPPOWER Power

7 . pca Telephone Bank Power, correlation

Principal components/correlation	Number of obs	=	168
	Number of comp.	=	3
	Trace	=	3
Rotation: (unrotated = principal)	Rho	=	1.0000

Component	Eigenvalue	Difference	Proportion	Cumulative
Comp1	1.64916	.631406	0.5497	0.5497
Comp2	1.01775	.684661	0.3393	0.8890
Comp3	.333091	.	0.1110	1.0000

Principal components (eigenvectors)

Variable	Comp1	Comp2	Comp3	Unexplained
Telephone	0.2224	0.9375	0.2675	0
Bank	0.7115	0.0315	-0.7020	0
Power	0.6666	-0.3465	0.6601	0

8 . predict index1
(score assumed)
(2 components skipped)

Scoring coefficients
sum of squares(column-loading) = 1

Variable	Comp1	Comp2	Comp3
Telephone	0.2224	0.9375	0.2675
Bank	0.7115	0.0315	-0.7020
Power	0.6666	-0.3465	0.6601

9 . gen y= ln(NSDPPerCapita)
(832 missing values generated)

10 . xtunitroot fisher y, dfuller lags(1)
(832 missing values generated)

Fisher-type unit-root test for y
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
Ha: At least one panel is stationary Number of periods = **14**

AR parameter: **Panel-specific** Asymptotics: **T -> Infinity**
Panel means: **Included**
Time trend: **Not included**
Drift term: **Not included** ADF regressions: **1 lag**

		Statistic	p-value
Inverse chi-squared(24)	P	4.2304	1.0000
Inverse normal	Z	4.0204	1.0000
Inverse logit t(64)	L*	3.9238	0.9999
Modified inv. chi-squared	Pm	-2.8535	0.9978

P statistic requires number of panels to be finite.
Other statistics are suitable for finite or infinite number of panels.

11 . xtunitroot fisher D.y, dfuller lags(1)
(844 missing values generated)

Fisher-type unit-root test for **D.y**
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
Ha: At least one panel is stationary Number of periods = **13**

AR parameter: **Panel-specific** Asymptotics: T -> **Infinity**
Panel means: **Included**
Time trend: **Not included**
Drift term: **Not included** ADF regressions: **1 lag**

		Statistic	p-value
Inverse chi-squared(24)	P	44.4489	0.0068
Inverse normal	Z	-3.3058	0.0005
Inverse logit t(64)	L*	-3.1316	0.0013
Modified inv. chi-squared	Pm	2.9515	0.0016

P statistic requires number of panels to be finite.
Other statistics are suitable for finite or infinite number of panels.

12 . xtunitroot fisher HDI, dfuller lags(1)
(832 missing values generated)

Fisher-type unit-root test for **HDI**
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
Ha: At least one panel is stationary Number of periods = **14**

AR parameter: **Panel-specific** Asymptotics: T -> **Infinity**
Panel means: **Included**
Time trend: **Not included**
Drift term: **Not included** ADF regressions: **1 lag**

		Statistic	p-value
Inverse chi-squared(24)	P	21.9447	0.5826
Inverse normal	Z	0.0612	0.5244
Inverse logit t(64)	L*	0.0302	0.5120
Modified inv. chi-squared	Pm	-0.2967	0.6166

P statistic requires number of panels to be finite.
Other statistics are suitable for finite or infinite number of panels.

13 . xtunitroot fisher D.HDI, dfuller lags(1)
(844 missing values generated)

Fisher-type unit-root test for **D.HDI**
Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
Ha: At least one panel is stationary Number of periods = **13**

AR parameter: **Panel-specific** Asymptotics: T -> Infinity
 Panel means: **Included**
 Time trend: **Not included**
 Drift term: **Not included** ADF regressions: 1 lag

		Statistic	p-value
Inverse chi-squared(24)	P	64.9498	0.0000
Inverse normal	Z	-5.0413	0.0000
Inverse logit t(64)	L*	-5.0221	0.0000
Modified inv. chi-squared	Pm	5.9106	0.0000

P statistic requires number of panels to be finite.
 Other statistics are suitable for finite or infinite number of panels.

14 . xtunitroot fisher index1, dfuller lags(1)
 (832 missing values generated)

Fisher-type unit-root test for **index1**
 Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
 Ha: At least one panel is stationary Number of periods = **14**

AR parameter: **Panel-specific** Asymptotics: T -> Infinity
 Panel means: **Included**
 Time trend: **Not included**
 Drift term: **Not included** ADF regressions: 1 lag

		Statistic	p-value
Inverse chi-squared(24)	P	17.3759	0.8322
Inverse normal	Z	1.4287	0.9234
Inverse logit t(64)	L*	1.6783	0.9509
Modified inv. chi-squared	Pm	-0.9561	0.8305

P statistic requires number of panels to be finite.
 Other statistics are suitable for finite or infinite number of panels.

15 . xtunitroot fisher D.index1, dfuller lags(1)
 (844 missing values generated)

Fisher-type unit-root test for **D.index1**
 Based on augmented Dickey-Fuller tests

H0: All panels contain unit roots Number of panels = **12**
 Ha: At least one panel is stationary Number of periods = **13**

AR parameter: **Panel-specific** Asymptotics: T -> Infinity
 Panel means: **Included**
 Time trend: **Not included**
 Drift term: **Not included** ADF regressions: 1 lag

		Statistic	p-value
Inverse chi-squared(24)	P	37.6843	0.0373
Inverse normal	Z	-1.7590	0.0393
Inverse logit t(64)	L*	-1.8668	0.0333
Modified inv. chi-squared	Pm	1.9752	0.0241

P statistic requires number of panels to be finite.

Other statistics are suitable for finite or infinite number of panels.

16 . reg y L.y HDI index1

Source	SS	df	MS	Number of obs	=	156
Model	33.7443886	3	11.2481295	F(3, 152)	=	5753.18
Residual	.297177738	152	.001955117	Prob > F	=	0.0000
				R-squared	=	0.9913
				Adj R-squared	=	0.9911
Total	34.0415664	155	.219623009	Root MSE	=	.04422

y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
y						
L1.	.9822153	.0146546	67.02	0.000	.9532622	1.011168
HDI	.14216	.1012749	1.40	0.162	-.0579283	.3422483
index1	.0028681	.0032611	0.88	0.381	-.0035749	.0093111
_cons	.1462282	.1172244	1.25	0.214	-.0853714	.3778278

17 . vif

Variable	VIF	1/VIF
y		
L1.	3.69	0.270654
HDI	3.03	0.329517
index1	1.42	0.705375
Mean VIF	2.72	

18 . estat hettest

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: Normal error terms
Variable: Fitted values of y

H0: Constant variance

chi2(1) = **2.30**
Prob > chi2 = **0.1295**

19 . gen y1=L.y
(844 missing values generated)

20 . xtserial y y1 HDI index1

Wooldridge test for autocorrelation in panel data
H0: no first-order autocorrelation
F(1, 11) = **9.454**
Prob > F = **0.0106**

21 . xtabond2 y L.y HDI index1, gmm(L.y HDI index1, lag(2 3) collapse) twostep robust
 Favoring space over speed. To switch, type or click on mata: mata set matafavor speed, perm.

Dynamic panel-data estimation, two-step system GMM

Group variable: state_id	Number of obs	=	156
Time variable : YEAR	Number of groups	=	12
Number of instruments = 10	Obs per group: min	=	13
Wald chi2(3) = 628289.68	avg	=	13.00
Prob > chi2 = 0.000	max	=	13

	y	Coefficient	Corrected std. err.	z	P> z	[95% conf. interval]	
	y						
	L1.	.9385457	.0374894	25.03	0.000	.8650678	1.012024
	HDI	.7013428	.3407404	2.06	0.040	.0335039	1.369182
	index1	.0361223	.0222861	1.62	0.105	-.0075577	.0798023
	_cons	.2526959	.31989	0.79	0.430	-.3742769	.8796688

Instruments for first differences equation

GMM-type (missing=0, separate instruments for each period unless collapsed)

L(2/3).(L.y HDI index1) collapsed

Instruments for levels equation

Standard

_cons

GMM-type (missing=0, separate instruments for each period unless collapsed)

DL.(L.y HDI index1) collapsed

Arellano-Bond test for AR(1) in first differences: z = **-2.63** Pr > z = **0.009**

Arellano-Bond test for AR(2) in first differences: z = **0.05** Pr > z = **0.959**

Sargan test of overid. restrictions: chi2(6) = **3.98** Prob > chi2 = **0.680**

(Not robust, but not weakened by many instruments.)

Hansen test of overid. restrictions: chi2(6) = **7.26** Prob > chi2 = **0.297**

(Robust, but weakened by many instruments.)

Difference-in-Hansen tests of exogeneity of instrument subsets:

GMM instruments for levels

Hansen test excluding group: chi2(3) = **2.79** Prob > chi2 = **0.425**

Difference (null H = exogenous): chi2(3) = **4.47** Prob > chi2 = **0.215**